

ECON 320 – Introduction to Mathematical Economics
Fall 2025

Instructor Information	
Instructor	Muhebullah Karimzada
Contact Information	✉ karimzada@unbc.ca ☎ Phone: TBA
Office Location & Hours	10-4522 ⌚ Mon: 10:30 am – 12:00 pm Tue: 1:30 pm – 3:00 pm Thu: 1:30 pm – 3:00 or by appointment
Teaching Assistant	✉ TBA

Course Information	
Lectures	Monday and Wednesday from 8:30 am to 9:50 pm in Room 8-161
Course Format	In-Person
Course Description	An introduction to the fundamental mathematical methods essential for the study of economic theory. Topics include comparative statics, rules of differentiation, implicit and general functional relationships, exponential and logarithmic functions, and techniques for optimizing single- and multi-variable functions. The course also introduces methods for analyzing economic dynamics, such as single-variable integration, first-order differential equations, and approaches to dynamic optimization.
Textbook	Required: <i>Alpha C Chiang & Kevin Wainwright. Fundamental Methods of Mathematical Economics, 4th edition, McGraw-Hill, 2005.</i> Supplementary Reading: <i>M. Hoy, J. Livernois, C. McKenna, R. Rees, and T. Stengos. Mathematics for Economics 3rd edition, The MIT Press, 2011.</i> <i>C. P. Simon and L. E. Blume. Mathematics for Economists, W. W. Norton & Company, 1994.</i>
Additional Resources	Additional resources (e.g. lecture slides, practice problem sets) are posted on Moodle. Students are responsible for downloading and saving course material on Moodle before the access to their courses is shut off.

Evaluation & Assessments

Learning in this course results primarily from lectures, in-class discussions, home readings, and practice problem sets. Students learning and progress in this course will be evaluated on individual basis. Grades for this course will be calculated as follows:

Assessment	Weight	Due Date
In-class quizzes	10%	Sep 17 th , 24 th , Oct 8 th , 20 th , 27 th and Nov 26 th .
Midterm exam 1	25%	October 1 st , 2025
Midterm exam 2	25%	Nov 5 th , 2025
Final Exam	40%	TBA
Total	100%	

Course schedule by week

Week	Date	Topic	Reading	Evaluation Due Dates
1	Sep 3 rd	Review of Sets and Functions	Ch. 2 (C&W)	
2	Sep 8 th & 10 th	Comparative Statics and the Concept of Derivative	Ch. 6 (C&W)	
3	Sep 15 th & 17 th	Rules of Differentiation and Their Use in Comparative Statics	Ch. 7 (C&W)	Quiz 1: Sep 17 th , 2025 Ch. 2 & 6
4	Sep 22 nd & 24 th	Comparative-Static Analysis of General Function Models	Ch. 8 (C&W)	Quiz 2: Sep 24 th , 2025 Ch. 7
5	Sep 29 th & Oct 1 st	Optimization of Single-Variable Functions	Ch. 9 (C&W)	Midterm exam 1: Oct 1 st , 2025 Ch. 2 & 6-8
6	Oct 6 th and 8 th	Exponential and Logarithmic Functions	Ch. 10 (C&W)	Quiz 3: Oct 8 th , 2025 Ch. 9
7	Oct 15 th	Optimization of Multi-Variable Functions	Ch. 11 (C&W)	
8-10	Oct 20 th to Nov 5 th	Optimization with Equality Constraints	Ch. 12 (C&W)	Quiz 4: Oct 20 th , 2025 - Ch. 10 Quiz 5: Oct 27 th , 2025 - Ch. 11 Midterm exam 2: Nov 5 th , 2025 Ch. 9-12
	Nov 10 th -14 th	UNBC Mid-Semester Break		

11	Nov 17 th & 19 th	Envelope Theorem and Duality	Ch. 13 (C&W)	
12	Nov 24 th & 26 th	Integration and Economic Dynamics	Ch. 14 (C&W)	Quiz 6: Nov 26 th , 2025 - Ch. 13
13	Dec 1 st and 3 rd	Final Exam Review		

Please Note:

1. For some chapters selected portions will be covered as a part of the course material.
2. If a student misses a lecture for any reason, it is student's responsibility to obtain class lecture notes and other relevant material.
3. There will be six in-class quizzes, with dates listed in the table above. Each quiz will be held during the last 30 minutes of the lecture. Only your top five scores will be counted, giving you one free pass. A missed quiz will automatically be considered your free pass. There will be no makeup quizzes.
4. There will be no make-up midterms.
5. If a midterm exam is missed, given proper documentation, 50% of the missed midterm weight will be transferred to the other midterm and the remaining 50% of the weight will be allocated to the final exam. Students must write at least one midterm exam.
6. Quizzes, midterms, and the final exam are strictly individual assessments and will be closed book.
7. Quizzes, midterms, and the final exam will feature a mix of question types, including short answer, analytical problems, numerical calculations, mathematical derivations, true/false, multiple choice, and fill-in-the-blank.
8. Your grades on quizzes and midterms will be posted on Moodle. Please immediately report to the instructor any discrepancy between the posted and actual grades.
9. Only the Registrar's Office can issue final grades. Hence, marks for the final exam will not be posted on Moodle.
10. You are encouraged to contact course instructor or course TA for any questions on the course material and outlines, evaluation scheme, and the final exam.

Artificial Intelligence (AI)

Use of generative AI is NOT allowed in this course, unless specifically authorized by the instructor. Please consult with the instructor if any clarification is needed. General guidance on the acceptability of using generative AI at UNBC can be found [here](#).

Plagiarism

Plagiarism is a misrepresentation of information that can result in a misattribution of credit, up to and including representing someone else's work as your own. When you incorporate the words, ideas, graphics, or other products from someone else's work into your projects, you must give credit by providing a citation and reference to the source work.

It is your responsibility to:

1. Familiarize yourself with and understand the information on [plagiarism](#) provided by the Academic Success Centre,
2. And the UNBC policy on Student Conduct, plagiarism and other academic offenses, as described in the UNBC Undergraduate calendar.

Grading System - Undergraduate Assessments

Grade Point	Letter Grade	Percentage	Definition/Standing
4.33	A+	90 -100%	Excellent
4.00	A	85-89.9%	
3.67	A-	80-84.9%	
3.33	B+	77-79.9%	Good
3.00	B	73-76.9%	
2.67	B-	70-72.9%	
2.33	C+	67-69.9%	Satisfactory
2.00	C	63-66.9%	
1.67	C-	60-62.9%	Marginal
1.33	D+	57-59.9%	
1.00	D	53-56.9%	
0.67	D-	50-62.9%	
0.00	F	0-49.9%	Failure

Student Services

University students often encounter setbacks from time to time that can impact academic performance. If you experience difficulties and need assistance, it's important to reach out. The following resources are available **free** to all students at UNBC and can help address mental, emotional, or physical health concerns:

General IT & Moodle Support

IT Services can assist you with accessing your UNBC account or Moodle, please email support@unbc.ca or visit their [Support for Students](#) page. If you have specific questions or errors while uploading assignments, using video, Kaltura etc. contact the Centre for Teaching Learning & Technology at ctlit@unbc.ca.

Academic Advising

For questions regarding course load, understanding your degree requirements or your GPA please reach out to Academic Advising located in the Enrollment Services Room 7-702.

Webpage: [Student Advising](#)

Phone: 250-960-6306

Email: advising@unbc.ca

The Academic Success Centre (ASC)

The Academic Success Centre provides students with free access to academic support services: Tutoring (by appointment, asynchronous online, or drop-in), personalized study skills assessments and peer-led course supports. ASC services are available in person (room 5-139) at the Prince George Campus and online

Webpage: [Academic Success Centre](#)

Phone: 250-960-6367 | Toll Free 1-888-440-3440

Email: asc@unbc.ca

Access Resource Centre (ARC)

The Access Resource Centre (ARC) provides services to students with documented health conditions and/or disabilities. ARC staff are available to meet with you to determine which academic accommodation can be put in place to support you in achieving their academic goals, provide referrals, and help advocate for you.

Webpage: [Access Resource Centre](#)

Phone: 250-960-5682 | Toll Free 1-888-960-5682

Email: arc@unbc.ca

UNBC Counselling Services

Offers counseling for all students as well as access to First Nations Counseling Services

Offers health and general services that support academic success and personal wellbeing of students

Webpage: [UNBC Counselling Services](#)

Phone: 250-960-6369

Email: wellness@unbc.ca

Crisis Centre for Northern BC (24/7)

If you are feeling overwhelmed, depressed, confused, suicidal, or just needing someone to listen to, the crisis line provides emotional support, suicide intervention and referral information. If you are unable to meet course expectations due to mental health reasons, please discuss your situation with your instructor, counsellor or/and an academic advisor.

Webpage: <https://crisis-centre.ca/>

Phone: Northern BC: [1-888-562-1214](tel:1-888-562-1214) | Youth Support: [1-888-564-8336](tel:1-888-564-8336) | Anywhere in BC: [1-800-784-2433](tel:1-800-784-2433)

UNBC SAFE app

UNBC SAFE features emergency contacts, services, information and safety tips. Emergency messaging can be sent to you via Push notifications when enabled. Download the App [here](#).

UNBC Athletics

Please note that students involved in University Athletics will be required to notify the instructor at least two weeks in advance of any official athletics commitments (e.g., games, travel) that conflict with due dates.

Food Security

We understand the cost of living is higher than ever, and it's hard to learn when you're hungry. The [NUGSS Food Bank](#) (located in the NUGSS office in Building 6) is open to all students.